

JAVA PROGRAMMING — PRACTICAL QUESTION PAPER

Laboratory Evaluation • Makeup Practical Session

Instructions: Write a complete Java program for each question. Use simple, clear code with proper comments. Questions move from basic to advanced across sessions.

SESSION 1 — PRINT STATEMENTS & VARIABLES

1. Print your name, class, and roll number using `System.out.println()`.
2. Print a welcome message for a shop, e.g., "Welcome to Sharma General Store".
3. Store your age in a variable and print it along with a message.
4. Store the price of a pen in a variable and print "The price of a pen is ₹10".
5. Store your favorite fruit's name and its price in two variables and print them together.
6. Print the number of students in a class using an `int` variable.

SESSION 2 — DATA TYPES & TYPE CASTING

7. Store the price of a chocolate as a `double` and print it.
8. Store the number of chairs in a classroom as an `int` and the average weight of a chair as a `double`.
9. Convert an `int` value of a person's age into a `double` and print both values.
10. Convert a `double` value of a bus fare into an `int` and print the result.
11. Store a book's price as a `float` and print it with two decimal points.
12. Store whether a shop is open or closed using a `boolean` variable and print it.

SESSION 3 — OPERATORS & EXPRESSIONS

13. Add the price of two items and print the total cost.
14. Subtract the discount amount from the original price of a shirt.
15. Multiply the number of pens by the price of one pen to get the total cost.
16. Divide the total marks by the number of subjects to find the average.
17. Use the modulus operator to check if a given roll number is odd or even.
18. Increase a product's price by 10% using arithmetic operators.

SESSION 4 — TAKING USER INPUT (SCANNER CLASS)

19. Take the user's name as input and print "Hello, [name]!".
20. Take two numbers as input and print their sum.
21. Take the price of an item and quantity as input, then print the total bill.
22. Take a student's marks as input and print them back.
23. Take the user's age as input and print it.
24. Take a city name as input and print "You live in [city]".

SESSION 5 — CONDITIONAL STATEMENTS: IF-ELSE & SWITCH

25. Check if a number entered by the user is positive or negative.
26. Check if a student has passed or failed based on marks (pass mark = 40).
27. Check if a person is eligible to vote (age 18 or above).
28. Using `switch`, print the name of the day for a number entered (1 = Monday, 2 = Tuesday, etc.).
29. Check if a number is even or odd using `if-else`.
30. Check if a triangle is valid based on three side lengths entered by the user.
31. Using `if-else`, give a grade (A, B, C, D) based on marks entered.

SESSION 6 — LOOPS: FOR, WHILE, DO-WHILE

32. Print numbers from 1 to 10 using a `for` loop.
33. Print the multiplication table of a number entered by the user.
34. Print all even numbers between 1 and 50 using a `while` loop.
35. Calculate the sum of numbers from 1 to n using a `for` loop.
36. Print "Good Morning" 5 times using a `do-while` loop.
37. Count how many days are left until the weekend using a loop (simple counter).
38. Print a countdown from 10 to 1 using a loop.

SESSION 7 — ARRAYS (1D & 2D)

39. Store the marks of 5 students in an array and print them.
40. Find the highest number in an array of 5 numbers.
41. Find the total and average of numbers stored in an array.
42. Store the names of 5 fruits in an array and print each one.
43. Create a 2D array to store a 3x3 seating plan and print it.
44. Count how many even numbers are present in an array.

SESSION 8 — STRINGS & STRING METHODS

45. Take a person's name as input and print its length using `length()`.
46. Check if a word entered by the user contains the letter 'a'.
47. Convert a given sentence to uppercase and lowercase.
48. Take a full name and print only the first name using `substring()`.
49. Compare two strings to check if they are the same using `equals()`.
50. Reverse a given word or sentence.

SESSION 9 — METHODS / FUNCTIONS

51. Write a method to add two numbers and call it from `main()`.
52. Write a method that finds the square of a number.
53. Write a method to check if a number is prime.
54. Write a method `greet()` that prints "Hello, welcome!" and call it.
55. Write a method that takes a person's age and returns whether they are a minor or adult.
56. Write two methods with the same name but different parameters (method overloading) to add two numbers and three numbers.

SESSION 10 — CLASSES, OBJECTS & CONSTRUCTORS

57. Create a class `Student` with fields `name` and `marks`, and print details using an object.
58. Create a class `Car` with fields `brand` and `color`, and create two car objects.
59. Create a class `Book` with a constructor that sets the book's title and author.
60. Create a class `Person` with fields `name` and `age`, and a method to display them.
61. Create a class `Rectangle` with fields `length` and `width`, and a method to calculate area.
62. Create a class `Fruit` and make an array of 3 fruit objects with name and price.

SESSION 11 — INHERITANCE

63. Create a class `Animal` with a method `sound()`, and a subclass `Dog` that overrides it.
64. Create a class `Vehicle` with a field `speed`, and a subclass `Bike` that adds a `brand` field.
65. Create a base class `Person` and a subclass `Student` that adds a `rollNumber` field.
66. Create a class `Employee` and a subclass `Manager` that adds an extra `bonus` field.
67. Use the `super` keyword to call a parent class constructor from a child class.

SESSION 12 — POLYMORPHISM, ABSTRACTION & INTERFACES

68. Create an interface `Shape` with a method `area()` and implement it in a class `Circle`.
69. Create an abstract class `Animal` with an abstract method `makeSound()`.
70. Show method overloading by creating two `add()` methods with different numbers of parameters.
71. Show method overriding by creating a parent class and child class with the same method name.
72. Create an interface `Payable` with a method `pay()` and implement it in a class `Employee`.

SESSION 13 — EXCEPTION HANDLING

73. Write a program that divides two numbers and handles the "divide by zero" error using `try-catch`.
74. Write a program that handles an `ArrayIndexOutOfBoundsException` when accessing an invalid array position.
75. Write a program that shows the use of `finally` block along with `try-catch`.
76. Write a program that throws a custom exception when a user enters a negative age.
77. Take a number as input and handle the error if the user enters text instead of a number.

SESSION 14 — COLLECTIONS: ARRAYLIST & HASHMAP

78. Create an `ArrayList` of fruit names and print all of them.
79. Add and remove an item from an `ArrayList` of tasks.
80. Create a `HashMap` to store student names and their marks, then print them.
81. Check if a particular item exists in an `ArrayList`.
82. Find the size of an `ArrayList` and print the first and last elements.

SESSION 15 — BONUS: FILE HANDLING

83. Write a program to write a simple message into a text file using `FileWriter`.
84. Write a program to read and display the content of a text file using `BufferedReader`.
85. Write a program that writes a list of items into a file, one per line.

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